Chemical Theory II

Statistical Thermodynamics

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The course is developed based on the
Cambridge chemistry courses MELT

Summary of last lecture

- Equipartition principle
  - Each "squared term" in the energy contributes ½RT to U (and ½R to Cv).
  - N.B. Vibrational degrees of freedom each contribute a whole RT.
  - Classical result.
  - Individual contributions appear when kT approaches the relevant energy.
- Contributions to U and Cv
  - Translation: $U = 3 \times$ ½RT at all temperatures.
  - Rotation: 2 or 3 times ½RT. Care needed for $T < \theta_{rot}$.
  - Vibration:
    - Equipartition often not applicable, since $T < \theta_{vib}$.
    - Taylor expansions of exponentials to obtain high T limits.
  - Electronic: only contributes if there are low-lying excited states.
  - Generally: $q = \text{constant} \times T^{\beta}$ leads to $U = \beta RT$.

Course outline

- Entropy

Entropy

- In ‘classical’ thermodynamics entropies are determined from measurements of heat capacities as a function of temperature; it is necessary to make such measurements down to as close to absolute zero as possible. This is not very easy to do, so the calculation of entropies using spectroscopic data and the methods of statistical thermodynamics is an attractive option. For gases, such calculations can often give more precise estimates of the entropy than are available from heat capacity measurements. There is an important contrast here between classical thermodynamics, which tells us how to compute entropies from heat capacities, and statistical thermodynamics which allows us to compute entropies *ab initio*.

Separating different contributions

- We saw before that it is possible to separate the entropy into contributions from different sets of energy levels:

$$S_{\text{trans}} = Nk \ln q_{\text{trans}} - Nk \ln N + Nk + NkT \left(\frac{\partial \ln q_{\text{trans}}}{\partial T} \right)_V ,$$

$$S_{\text{rot}} = Nk \ln q_{\text{rot}} + NkT \left(\frac{\partial \ln q_{\text{rot}}}{\partial T} \right)_V ;$$

  there are similar expressions to $\boldsymbol{S_{rot}}$ for the vibrational and electronic contributions.

- It is sometimes useful to rewrite these expressions to include the internal energy, ***U***; for each contribution to the internal energy the expression for ***U*** is the same:

$$U_{\text{trans}} = NkT^2 \left(\frac{\partial \ln q_{\text{trans}}}{\partial T} \right)_V \qquad U_{\text{rot}} = NkT^2 \left(\frac{\partial \ln q_{\text{rot}}}{\partial T} \right)_V \quad \text{etc.}$$

Separating different contributions

- It follows that

$$\frac{U_{\text{trans}}}{T} = NkT\left(\frac{\partial \ln q_{\text{trans}}}{\partial T}\right)_V$$

which can be inserted into the right-hand side of the first equation in the last slide

$$S_{\text{trans}} = Nk \ln q_{\text{trans}} - Nk \ln N + Nk + \frac{U_{\text{trans}}}{T}$$

and likewise in the second equation to give

$$S_{\text{rot}} = Nk \ln q_{\text{rot}} + \frac{U_{\text{rot}}}{T}.$$

- In the same way, the vibrational contribution is

$$S_{\text{vib}} = Nk \ln q_{\text{vib}} + \frac{U_{\text{vib}}}{T}.$$

Note that when using this equation it is important to be consistent in the choice of energy zero for computing $\boldsymbol{q_{vib}}$ and $\boldsymbol{U_{vib}}$.

Translational entropy

- The translational partition function in three dimensions is given:

$$q_{\text{trans}} = \left(\frac{2\pi mkT}{h^2}\right)^{\frac{3}{2}} V$$

and we have seen that this leads to an internal energy of ***3/2 NkT***. It is therefore convenient to obtain the translational entropy

$$\begin{aligned} S_{\text{trans}} &= Nk \ln\left\{\left(\frac{2\pi mkT}{h^2}\right)^{\frac{3}{2}} V\right\} - Nk\ln N + Nk + \frac{\frac{3}{2}NkT}{T} \\ &= Nk\left[\ln\left\{\left(\frac{2\pi mkT}{h^2}\right)^{\frac{3}{2}} V\right\} - \ln N + \frac{5}{2}\right]. \end{aligned}$$

- This is often called the Sackur–Tetrode equation. Note that the expression we have used for the partition function was derived under the assumption that the energy levels are closely spaced compared to ***kT***. Under these circumstances we can simply write down the value for $\boldsymbol{U_{trans}}$ as ***3/2 NkT*** using the equipartition principle; it is not necessary to compute it from $\boldsymbol{q_{trans}}$.

Translational entropy

- The molar entropy is simply found by setting N = N_A:

$$S_{\text{trans,m}} = R\left[\ln\left\{\left(\frac{2\pi mkT}{h^2}\right)^{\frac{3}{2}} V_{\text{m}}\right\} - \ln N_{\text{A}} + \frac{5}{2}\right]$$

- In this expression the molar volume, $\boldsymbol{V_m}$, can be calculated using the ideal gas equation, $\boldsymbol{pV_m = RT}$.
- N.B. Mind the units when doing the calculations.

Rotational entropy

- If the temperature is high enough that the rotational energy levels are spaced closely compared to ***kT***, it was shown before that the rotational partition function for a diatomic is given by

$$q_{\mathrm{rot}} = \frac{T}{\sigma\theta_{\mathrm{rot}}}.$$

- Under these conditions the equipartition principle applies and the internal energy, ***U_{rot}***, is ***NkT***. Hence, the rotational entropy can be written

$$\begin{aligned} S_{\mathrm{rot}} &= Nk\ln q_{\mathrm{rot}} + \frac{U_{\mathrm{rot}}}{T} \\ &= Nk\ln\frac{T}{\sigma\theta_{\mathrm{rot}}} + \frac{NkT}{T} = Nk\left(\ln\left\{\frac{T}{\sigma\theta_{\mathrm{rot}}}\right\} + 1\right) \end{aligned}$$

  The same expression can be used for linear polyatomic molecules.

- For non-linear molecules we need to remember that in the high-temperature limit the internal energy due to rotation is ***3 × 1/2 NkT = 3/2 NkT*** as there are three rotational degrees of freedom (in contrast to the two degrees of rotational freedom for a linear molecule). The expression for the partition function was given in previous lectures.

Vibrational entropy

- The partition function for a harmonic oscillator (taking the ground state to have zero energy) is given

$$q'_{\mathrm{vib}} = \frac{1}{1 - \exp\left(\frac{-\theta_{\mathrm{vib}}}{T}\right)}.$$

- The corresponding internal energy is,

$$U'_{\mathrm{vib}} = \frac{Nk\theta_{\mathrm{vib}}}{\exp\left(\frac{\theta_{\mathrm{vib}}}{T}\right) - 1}.$$

- The vibrational entropy is best calculated by finding ***q′***

  separately and then substituting these into

$$S_{\mathrm{vib}} = Nk \ln q'_{\mathrm{vib}} + \frac{U'_{\mathrm{vib}}}{T}$$

  For a polyatomic molecule, each normal mode makes a separate contribution to the entropy.

- If, as is often the case, $\boldsymbol{\theta_{vib} >> T}$, $\boldsymbol{q'_{vib} = 1}$ and $\boldsymbol{U'_{vib} = 0}$; the vibrational contribution to the entropy is zero. This is what we expect for the case when all the particles are in the ground state.

Electronic entropy

- It is usually the case that only the electronic ground state is occupied and that, if the energy of this state is taken to be zero, the electronic partition function is *just* $\boldsymbol{g_0}$*, the degeneracy of the ground state. The electronic contribution to the internal energy,* $\boldsymbol{U'_{elec}}$*, is zero* so the electronic contribution to the entropy is given by

$$S_{\text{elec}} = Nk \ln q'_{\text{elec}} + \frac{U'_{\text{elec}}}{T} = Nk \ln g_0 .$$

- Note that we were careful to choose the same energy zero for computing ***q*** and ***U***.

Comparison with calorimetric values

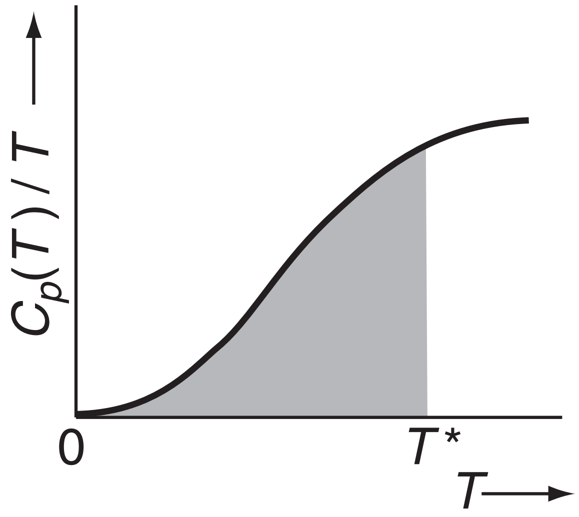
- In classical thermodynamics, the entropy at some temperature T*, S(T*), is determined from the integral

$$S(T^\star) = \int_0^{T^\star} \frac{C_p(T)}{T}\,\mathrm{d}T$$

- where ***$C_p(T)$*** is the constant pressure molar heat capacity at temperature ***T***. Since ***$C_p(T)$*** varies strongly with ***T*** this integral has to be evaluated graphically. The heat capacity is measured at many temperatures from ***T**** down to absolute zero and then these data are plotted as ***$C_p(T)/T$*** against ***T*** ; the area under the graph is ***S(T*)***.

Comparison with calorimetric values

- In practice, it is not possible to make measurements down to absolute zero – in fact, reaching anything lower that the temperature of liquid helium (about 4 K) is quite difficult. It is therefore necessary to extrapolate the graph from the lowest achievable temperature down to zero.



$S(T^\star)$ is the shaded area.

- Luckily, *at these temperatures most insulating solids obey the Debye law, which is that* $\boldsymbol{C_p(T) = aT^3}$, where a is a constant characteristic of the material. The value of a is determined from the lowest temperature measurements which can conveniently be made, and then this formula is used to extrapolate the graph to absolute zero.

Comparison with calorimetric values

- In addition, if the material undergoes any phase transitions (e.g. gas to liquid, liquid to solid), the entropy changes of these needs to be taken into account. It can be shown that the entropy of a phase change, ΔS_{pc} is given by

$$\Delta S_{\mathrm{pc}} = \frac{\Delta H_{\mathrm{pc}}}{T_{\mathrm{pc}}}$$

  where ΔH_{pc} is the enthalpy of the phase change taking place at temperature T_{pc}. The apparatus used to measure the heat capacity is also capable of measuring the enthalpy of any phase changes.
- The full expression for computing S(T*) is therefore

$$S(T^\star) = \int_0^{T^\star} \frac{C_p(T)}{T}\,\mathrm{d}T + \sum_{\text{phase changes}} \frac{\Delta H_{\mathrm{pc}}}{T_{\mathrm{pc}}}$$

- As you can imagine, making these kinds of measurements is both very difficult and tedious. Nevertheless, they have been made for a vast number of compounds, and it is such measurements which lead to the values of the entropies you will find in tables.

Comparison with calorimetric values

- As we have discussed, for simple gases we can compute the entropy rather straightforwardly using the methods of statistical thermodynamics. In a number of cases the values so obtained have been compared with those from thermochemical methods, and excellent agreement has been found, as is illustrated in the table (standard molar entropies of the gases, given in J K^{-1} mol^{-1} at 298 K, data from McClelland)

molecule	O_2	N_2	Cl_2	HCl	Ar	CO_2
computed	205.1	191.5	223.0	186.8	154.7	213.7
thermochemical	205.4	192.2	223.9	186.1	154.6	213.8

- However, in a number of well-attested cases there are discrepancies between the computed and thermochemical values. In all such cases the calculated value exceeds the thermochemical value. For example, *for carbon monoxide the calculated value is 197.9 J K^{-1} mol^{-1}, but the thermochemical value is 193.4 J K^{-1} mol^{-1}*. *The discrepancy is too great to be assigned to experimental error*.

Comparison with calorimetric values

- What is thought to be going on here is that the thermochemical measurement is too low as a result of a phase transition not being included in the measurement. In the case of CO, it is thought that when the solid forms the molecules are oriented randomly e.g.
  C–O O–C O–C C–O O–C C–O C–O O–C
- This ‘freezes in’ some randomness into the crystal which, in principle, ought to be eliminated as the temperature drops so as to give a perfectly ordered arrangement at absolute zero e.g.
  C–O C–O C–O C–O C–O C–O C–O C–O
- The problem is that if the energy difference between these two arrangements is very small (which it appears to be), then the phase transition will only take place at a very low temperature. Furthermore, by the time we reach this temperature the transition may take place so slowly that it is never observed in any practical experiment. *The enthalpy change of the phase transition is therefore not measured, and hence our thermochemical value of the entropy is too lo*w.

Comparison with calorimetric values

- For simple molecules it is possible to make an estimate of this frozen in or residual entropy, as it is often known. If the CO molecule can adopt two different orientations with equal probability, then the number of possible microstates for ***N*** molecules is simply $\boldsymbol{2^N}$. The entropy can therefore be computed as

$$
\begin{aligned}
S &= k \ln \Omega \\
&= k \ln 2^N \\
&= Nk \ln 2.
\end{aligned}
$$

- *The molar residual entropy is therefore* ***R ln 2*** *(i.e. with $N = N_A$), which is 5.8 J K^{-1} mol^{-1}*. The difference in entropy between the calculated and calorimetric value is 197.9 – 193.4 = 4.5 J K^{-1} mol^{-1}, which is tolerably close to our estimate of the residual entropy.
- For other molecules which are nearly symmetrical, such residual entropies have been detected.

Isotopes

- A sample of Cl_2 gas is a mixture of the isotopomers $^{35}Cl-{}^{35}Cl$, $^{37}Cl-{}^{37}Cl$ and $^{35}Cl-{}^{37}Cl$. The mixing of these isotopomers has associated with it an increase in the entropy. When the gas freezes this mixing remains right down to the lowest temperatures simply because the energy associated with the separation of the mixture is so miniscule that the phase transition can never be realised experimentally. As a result, there is again some frozen-in randomness.
- *If we want our calorimetrically determined value of the entropy to be equal to the calculated value, then we must not include the entropy of mixing in the latter. This essentially boils down to ignoring the contribution made by the fact that isotopes are present. For any practical calculation on a reaction or equilibrium this will not have any effect since the mixing of isotopes is present in both reactants and products, and so cancels out*.

Course outline

- Nuclear spin statistics

Nuclear spin statistics

- Previously we introduced, without justification, *the symmetry parameter, **σ***, which was needed for the correct calculation of the rotational partition function.
- In this section we will discover that *this factor originates from a consideration of the symmetry of the molecular wavefunction with respect to exchange of nuclei*. You have already come across another example of this kind of symmetry in the *Pauli Principle* which requires the molecular wavefunction to be anti-symmetric with respect to exchange of electrons.

Nuclear spin statistics

- We will also show that *these symmetry considerations have wider consequences than a mere factor of two in calculating a partition function.*
- For example, we will be able to explain why it is in the rotational Raman spectrum of $^{16}O_2$ only lines originating from rotational states with odd ***J*** are seen, whereas in $^{12}C^{16}O_2$ only lines from even states are seen. We will also see that there are in fact two forms of $^{1}H_2$, *ortho* and *para*, which do not interconvert under ordinary circumstances, which can be separated and which have different physical properties.

Nuclear spin statistics

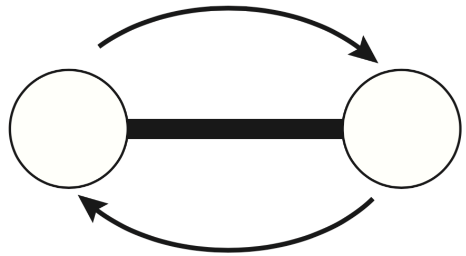
- The starting point for the discussion is *the factorization of the molecular wavefunction into contributions from different kinds of motion:*

$$\Psi = \psi_{\mathrm{electronic}}\psi_{\mathrm{vibrational}}\psi_{\mathrm{rotational}}\psi_{\mathrm{nuclearspin}}$$

- This is *the Born–Oppenheimer separation* which you have encountered before. A wavefunction describing the nuclear spins has been included for this discussion. *Normally we do not need to consider this when thinking about molecular spectra as the influence of the nuclear spins is very small and usually only detectable at the very highest resolution. However, for the present discussion the nuclear spin wavefunction is crucial.*

Nuclear spin statistics

- If we swap two identical nuclei (for example the two ^1H nuclei in $^1H_2$) then we know that the molecular wavefunction must either remain the same or, at most, change sign. *Remember that a sign change is allowed as ultimately all physical observables depend on Ψ^2 which is unaffected by a sign change in Ψ.* It turns out that nuclei can be classified into two types with respect to exchange: *bosons, which have the property that the wavefunction remains the same, and fermions which have the property that the wavefunction changes sign.*
- It is on this apparently simple statement that the whole of this section rests. We first need to look at nuclear spin wavefunctions and their symmetry properties, and then at exactly how this exchange of identical spins is brought about.



$$\Psi \xrightarrow[\text{bosons}]{\text{exchange}} \Psi$$

$$\Psi \xrightarrow[\text{fermions}]{\text{exchange}} -\Psi$$

Exchanging two identical nuclei either leads to no change in the wavefunction if the nuclei are bosons, or a sign change if the nuclei are fermions.

Nuclear spin wavefunctions

- A nucleus is characterised by *a nuclear spin quantum number, I,* which can have integer or half-integer values. Associated with the spin is *a magnetic moment which can interact with a magnetic field thereby giving rise to* ***(2I + 1)*** *energy levels which are characterised by the quantum number* $\boldsymbol{m_I}$; $\boldsymbol{m_I}$ takes values between $-I$ and I in integer steps. It is transitions between these energy levels which are observed in NMR spectroscopy. *In the absence of a magnetic field these levels are degenerate; we say that the spin I has associated with it a degeneracy of* ***(2I + 1)****.*

Nuclear spin wavefunctions

- In a diatomic, the first nucleus has **($2I_1$+ 1)** levels and the second has **($2I_2$+ 1)**. *There is only a very weak interaction between the nuclei, so to an excellent approximation we can specify the* ***m_I*** *value for each of the nuclei separately.* Any value of for the first spin can go with any value of ***m_I*** for the second, so that *there are a total of* ***($2I_1$+ 1)×($2I_2$+ 1)*** *levels for the diatomic*.

- To take a concrete example, consider $^{1}H^{19}F$; both nuclei have ***I* = 1/2** and so there are two values of ***m_I*: +1/2, -1/2**; it is usual to denote these as ***α*** and ***β***. There are a total of four nuclear spin wavefunctions for the diatomic and these can be written as

$$\alpha(\mathrm{H})\alpha(\mathrm{F}), \quad \alpha(\mathrm{H})\beta(\mathrm{F}), \quad \beta(\mathrm{H})\alpha(\mathrm{F}), \quad \beta(\mathrm{H})\beta(\mathrm{F})$$

where ***α*(H)** means that the ^{1}H nucleus is in state ***α*** and so on.

Nuclear spin wavefunctions

- If we consider a homonuclear diatomic the situation is a little more complex as the two nuclei are identical. Let us suppose that we label the two nuclei 1 and 2. *For a wavefunction to be valid it must either remain the same or at most change sign under interchange of the labels 1 and 2 .* For it to do any more would be physically unacceptable *as the nuclei are in fact identical and the labels are merely convenient fictions.*

- So, the following wavefunction is acceptable as there is no sign change on swapping the labels

$$\alpha(1)\alpha(2) \xrightarrow{\text{swap 1 and 2}} \alpha(2)\alpha(1) \equiv \alpha(1)\alpha(2)$$

whereas the next wavefunction is unacceptable as swapping the labels leads to a completely different function.

$$\alpha(1)\beta(2) \xrightarrow{\text{swap 1 and 2}} \alpha(2)\beta(1)$$

Nuclear spin wavefunctions

- In fact there are four wavefunctions which satisfy these requirements; three are symmetric with respect to swapping the labels:

$$\alpha(1)\alpha(2)$$
$$\left(1/\sqrt{2}\right)\left[\alpha(1)\beta(2)+\beta(1)\alpha(2)\right]$$
$$\beta(1)\beta(2)$$

  and one is anti-symmetric with respect to the same process:

$$\left(1/\sqrt{2}\right)\left[\alpha(1)\beta(2)-\beta(1)\alpha(2)\right]$$

- The factors of $1/\sqrt{2}$ are needed for normalization purposes.

Nuclear spin wavefunctions

$$\alpha(1)\alpha(2)$$

$$\left(1/\sqrt{2}\right)[\alpha(1)\beta(2)+\beta(1)\alpha(2)] \qquad \left(1/\sqrt{2}\right)[\alpha(1)\beta(2)-\beta(1)\alpha(2)]$$

$$\beta(1)\beta(2)$$

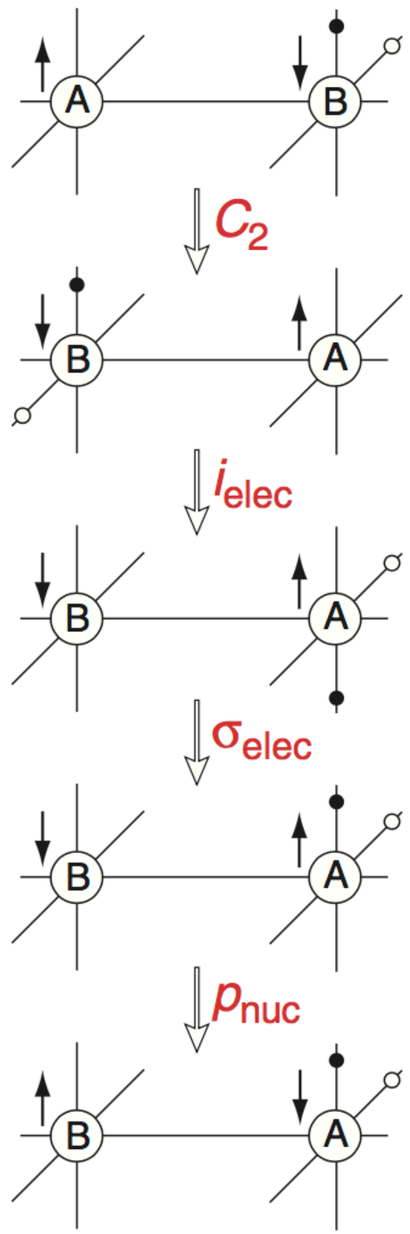
- These wavefunctions above should be familiar as being analogous to those for two electrons, say in the helium atom. In the first case they correspond to the three sublevels of a triplet **(*S* = 1, *M*$_s$= 1, 0, –1)** and in the second case to the one sublevel of a singlet **(*S* = 0, *M*$_s$=0)**.
- *In general, for a homonuclear diatomic, it can be shown that there are **$I(2I + 1)$** nuclear spin states which are anti-symmetric with respect to swapping the labels and **$(I + 1)(2I + 1)$** symmetric states. The total number of spin states is, as expected, **$(2I + 1)^2$**.*

How to swap the nuclei

- Recall that *the important thing we need to consider is what happens to the wavefunction when we swap the two nuclei*. However, ‘swapping the nuclei’ is not quite as simple as it sounds, since *what we want to do is swap the nuclei and leave everything else the same*.
- Just rotating the molecule by 180° does swap the nuclei but it also affects other parts of the molecular wavefunction (such as the electronic and rotational part), so we have not achieved what we wanted.
- In fact a more complex series of operations is needed to swap the nuclei and leave everything else the same; these are shown in the figure opposite.

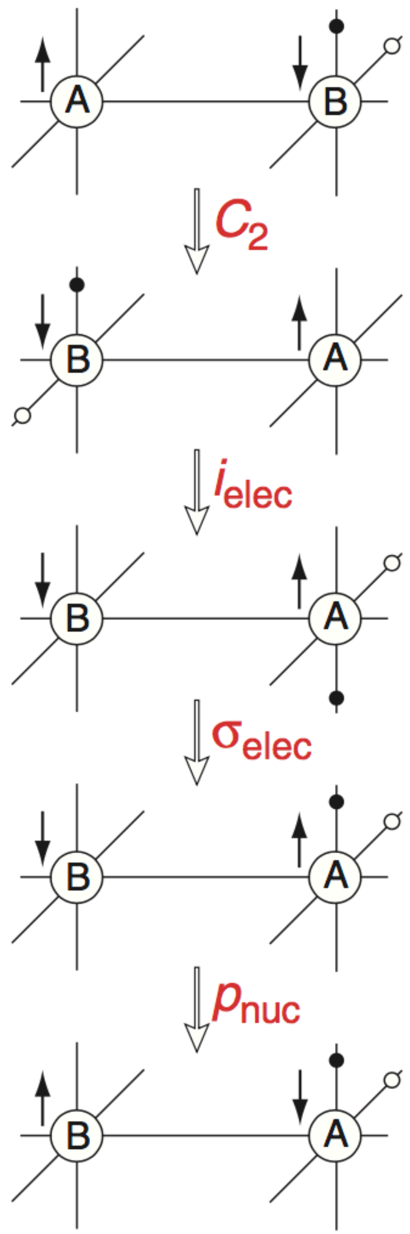
How to swap the nuclei

- *The diagram is fairly involved as we have to keep track of everything.* The letters A and B label the nuclei (they are of course identical, but we need the labels to keep track of them). Also shown by means of the arrows next to the nuclei are the spin states; arbitrarily we have shown A with spin up and B with spin down. Two 'typical' electrons are also shown, one by a small filled circle and one by an open circle.



How to swap the nuclei

- The sequence of operations is:
  - Rotate the whole molecule by 180° about an axis perpendicular to the internuclear axis (in Group Theory notation, $\boldsymbol{C_2}$). This swaps the nuclei, the spin states and moves the electrons.
  - Invert just the electrons though the centre of inversion ($\boldsymbol{i_{elec}}$).
  - Reflect just the electrons in a mirror plane containing the two atoms ($\boldsymbol{\sigma_{elec}}$).
  - Finally, an operation denoted $\boldsymbol{p_{nuc}}$ which permutes the nuclear spin states. This means that if spin A is up and spin B is down the result of $\boldsymbol{p_{nuc}}$ would be for spin A to be down and B up.



Effect on the molecular wavefunction

- We now have to consider, in turn, the effect of the four operations on the molecular wavefunction.
  - $\boldsymbol{C_2}$ of the whole molecule
  - Reflection of the electrons through the centre of inversion ($\boldsymbol{i_{\mathrm{elec}}}$)
  - Reflection of the electrons through a plane containing the two atoms ($\boldsymbol{\sigma_{\mathrm{elec}}}$)
  - Permutation of the nuclear spin states ($\boldsymbol{p_{\mathrm{nuc}}}$)

C_2 of the whole molecule

$$\Psi = \psi_{\text{electronic}}\psi_{\text{vibrational}}\psi_{\text{rotational}}\psi_{\text{nuclearspin}}$$

- *The electronic wavefunction is unaffected by this rotation as the wavefunction depends on 'where' the electrons are relative to the nuclei*; turning the whole molecule around (nuclei and electrons) does not alter the spatial relationship between the electrons and the nuclei, and *so $\boldsymbol{\psi_{elec}}$ is unaffected.*
- *For diatomics the vibrational wavefunction is also unaffected by this rotation of the whole molecule*. Recall that the vibrational wavefunction is expressed in terms of a co-ordinate which is the displacement from the equilibrium position. A positive displacement is the still a positive displacement, even if we turn the entire molecule through 180°. *Thus $\boldsymbol{\psi_{vib}}$ is unaffected.*

C_2 of the whole molecule

$$\Psi = \psi_{\text{electronic}}\psi_{\text{vibrational}}\psi_{\text{rotational}}\psi_{\text{nuclearspin}}$$

- *The rotational wavefunctions are affected.* Recall that for the rigid rotor the wavefunctions are the spherical harmonics, which are the same as the angular parts of the hydrogen orbitals. The wavefunction with $\boldsymbol{J = 0}$ is like an *s* orbital, and so is invariant to a rotation by 180°. In contrast, if $\boldsymbol{J = 1}$ the wavefunction is like a *p* orbital and so rotation by 180° results in a sign change. *In general, for even* $\boldsymbol{J}$ $\boldsymbol{\psi_{rot}}$ *is unchanged by* $\boldsymbol{C_2}$ *whereas for odd J there is a sign change.*
- *The nuclear spin wavefunction is unaffected by the rotation*; spin A is still up and spin B is still down.

Reflection of the electrons through the centre of inversion

$$\Psi = \psi_{\text{electronic}}\psi_{\text{vibrational}}\psi_{\text{rotational}}\psi_{\text{nuclearspin}}$$

- *This operation,* $\boldsymbol{i}_{\mathbf{elec}}$*, clearly only affects* $\boldsymbol{\psi}_{\mathbf{elec}}$. Knowledge of the molecular term symbol will enable us to determine the effect of $\boldsymbol{i}_{\mathbf{elec}}$. If the term symbol has *g* (gerade) symmetry, then $\boldsymbol{\psi}_{\mathbf{elec}}$ is unaffected by $\boldsymbol{i}_{\mathbf{elec}}$; if the term symbol has *u* (ungerade) symmetry, then $\boldsymbol{\psi}_{\mathbf{elec}}$ changes sign under $\boldsymbol{i}_{\mathbf{elec}}$.

Reflection of the electrons through a plane containing the two atoms

$$\Psi = \psi_{\text{electronic}}\psi_{\text{vibrational}}\psi_{\text{rotational}}\psi_{\text{nuclearspin}}$$

- *This operation, $\boldsymbol{\sigma}_{\mathbf{elec}}$, clearly only affects $\boldsymbol{\psi}_{\mathbf{elec}}$.* The superscript + or – from the molecular term symbol will enable us to determine the effect of $\boldsymbol{\sigma}_{\mathbf{elec}}$. If the superscript is +, then $\boldsymbol{\psi}_{\mathbf{elec}}$ is unaffected by $\boldsymbol{\sigma}_{\mathbf{elec}}$; if the superscript is – , then $\boldsymbol{\psi}_{\mathbf{elec}}$ changes sign under $\boldsymbol{\sigma}_{\mathbf{elec}}$.

Permutation of the nuclear spin states

- *Under this operation (**p_{nuc}**), it is clear that symmetric nuclear spin states* (as defined in earlier slides) *will not change sign whereas anti-symmetric ones will.* For example:

$$\text{symm:}\quad [\alpha(1)\beta(2)+\beta(1)\alpha(2)] \xrightarrow{p_{\text{nuc}}} [\beta(1)\alpha(2)+\alpha(1)\beta(2)]$$

$$\text{anti-symm:}\quad [\alpha(1)\beta(2)-\beta(1)\alpha(2)] \xrightarrow{p_{\text{nuc}}} [\beta(1)\alpha(2)-\alpha(1)\beta(2)]$$